GigaWhat Energy

GigaWhat Gas
Controlled procedure

PR-GAS-052 Safe Excavation Near Live Gas Mains

Field	Value
Document ID	PR-GAS-052
Version	4
Effective from	2024-02-05
Review due	2027-02-05
Owner	Head of Gas Network Operations
Approved by	Gas Safety Manager
Applies to	All GigaWhat Gas sites: low, medium and intermediate pressure mains

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

1 Purpose

This procedure sets out how to excavate safely near live GigaWhat Energy gas mains, so that mains are found before they are struck and are protected while exposed.

2 Scope

This procedure applies to all excavation within 5 m of a live LP, MP or IP main by GigaWhat Energy staff and contractors, including bar holing and trial holes. It applies to excavations near MAIN-IP-01, MAIN-MP-03, MAIN-LP-17 and MAIN-LP-22, and to all other mains on the network.

3 References

- PR-GAS-021 Permit-to-Work for Gas Operations
- PR-GAS-003 Response to Reported Gas Escapes
- PR-GAS-040 Working in Hazardous Areas (DSEAR Zones)
- PR-CORP-001 Reporting Incidents and Near Misses
- Pipelines Safety Regulations 1996
- Gas Safety (Management) Regulations 1996 and HSE guidance L80
- Management of Health and Safety at Work Regulations 1999

4 Definitions

Cable avoidance tool (CAT): A hand-held locator that detects buried metallic services by the signals they carry.

Signal generator: A transmitter used with the CAT to put a traceable signal on a metallic main or tracer wire.

Hand-dig zone: The area around a main where only hand tools may be used.

Trial hole: A hole dug by hand to expose a main and confirm its position and depth.

5 Responsibilities

Head of Gas Network Operations: Makes sure excavation teams are trained in this procedure and in the use of the CAT.

Authorised Person (Gas): Issues permits for mechanical excavation within 3 m of MAIN-IP-01 and advises on work near cast iron mains.

Excavation supervisor (a Competent Person (Pressure Systems) or contractor supervisor): Obtains plans, locates and marks mains, and controls the excavation.

Gas Network Controller: Receives reports of any damage to a main.

6 Procedure

6.1 Plans and planning

6.1.1 Obtain GigaWhat Energy gas mains plans, and plans from other utilities, dated no more than 30 days before the start of work.

6.1.2 Treat plans as a guide only. Mains may not be where the plans show, and depths are not guaranteed.

6.1.3 Mark the expected line of every main on site with yellow paint before digging.

6.1.4 For any work within 3 m of MAIN-IP-01, obtain a permit-to-work under PR-GAS-021 before any mechanical excavation.

6.2 Locating mains with the CAT

> CAUTION: Polyethylene mains such as MAIN-MP-03 and MAIN-LP-17 cannot be found with a CAT unless a tracer wire was laid with them. Absence of a signal does not mean there is no main.

6.2.1 Check the CAT and signal generator are within calibration and pass their self-test.

6.2.2 Sweep the whole work area in power mode and radio mode, then in signal generator mode.

6.2.3 Use the signal generator on metallic mains (MAIN-IP-01 steel, MAIN-LP-22 cast iron) and on any tracer wire.

6.2.4 Mark every located service on the surface and compare it with the plans.

6.2.5 Re-scan the excavation every 300 mm of depth as digging progresses.

6.3 Trial holes and hand-dig zones

> WARNING: Striking a live main with an excavator can release gas at high pressure and cause fire, explosion and serious injury. Hand digging near mains is mandatory.

6.3.1 Dig trial holes by hand to expose and confirm the position and depth of each main before any mechanical excavation within the work area.

6.3.2 Hand dig within 0.5 m either side of the confirmed line of any LP or MP main.

6.3.3 Hand dig within 1.5 m either side of the confirmed line of MAIN-IP-01, and keep mechanical excavators at least 3 m away unless the permit allows otherwise.

6.3.4 Use spades and shovels with careful, angled strokes. Do not use picks, forks, pins or bars to break ground over a main.

6.3.5 Do not use breakers within 0.5 m of any main.

6.4 Working near exposed mains

6.4.1 Support exposed PE and steel mains at intervals of no more than 1.5 m, and never use a main as a step or lifting point.

6.4.2 Do not undermine more than 1 m of cast iron main, such as MAIN-LP-22, without support and the advice of the Authorised Person (Gas). Cast iron fractures easily.

6.4.3 Keep heavy plant and spoil at least 1 m back from the edge of an excavation containing a main.

6.4.4 Keep these minimum clearances when laying other plant near a gas main: 300 mm parallel and 150 mm crossing for LP and MP mains; 600 mm parallel and 300 mm crossing for MAIN-IP-01.

6.4.5 Backfill with at least 150 mm of fine fill free of stones around the main, and lay marker tape 300 mm above it.

6.5 Damage to a main

6.5.1 Report any damage to a main to the Gas Network Controller, even if no gas is escaping, including scrapes to the coating of MAIN-IP-01 and scores on PE mains.

6.5.2 Do not backfill a damaged main until it has been inspected. PE mains with scores deeper than 10% of the wall thickness must be repaired.

6.5.3 Report every strike or near miss under PR-CORP-001.

6.6 Emergency

6.6.1 If gas is escaping, stop all engines, do not try to stop the gas, and get everyone out of the excavation and upwind.

6.6.2 Keep the public and vehicles at least 25 m away, or 50 m for MAIN-IP-01, and keep ignition sources away.

6.6.3 Report the escape to the Gas Network Controller for action under PR-GAS-003, or call the National Gas Emergency Service on 0800 111 999.

6.6.4 Call 999 for fire or injury.

7 Records

The excavation supervisor keeps the plans used, the CAT sweep record and trial hole depths with the job record. Damage reports are logged in the work management system and kept for the life of the main.

8 Revision history

Version	Date	Change
1	2012-02-06	First issue.
2	2016-02-01	Added hand-dig zones and minimum clearances.
3	2020-02-03	Added warning that PE mains cannot be found with a CAT without tracer wire.
4	2024-02-05	Hand-dig zone for MAIN-IP-01 increased from 1.0 m to 1.5 m. Added cast iron undermining limit and 300 mm re-scan interval.